

SymPy Bug Card

Bug 20: conditional uniform density keeps old support

Task. Compute the density of a uniform random variable conditioned on being larger than $1/2$.

$$Y = \text{given}(U, U > 1/2), \quad U \sim \text{Uniform}(0, 1).$$

SymPy's answer.

$$\text{density}(Y)(x) = 2 \cdot \mathbf{1}_{[0,1]}(x).$$

Correct answer.

$$\text{density}(Y)(x) = 2 \cdot \mathbf{1}_{(1/2,1]}(x).$$

Explanation. Conditioning on $U > 1/2$ must both divide by $P(U > 1/2) = 1/2$ and restrict the support to $(1/2, 1]$. Note that SymPy's returned density has total mass 2.

Likely root cause. The diagnosis identifies `ContinuousPSpace.conditional_space` in `sympy/stats/crv.py` as storing only the renormalized original PDF, while `compute_density` returns it unchanged when no variables are marginalized. This is a new failure mode with no public precedent.

Artifact (GitHub).

[bug-report/bug-020-conditional-uniform-density-keeps-old-support/](#)